Lam Fung Academy · LF Academy

Primary 5 · 7th Lesson · Student Handout

Fractions with Different Denominators: Comparison, Addition and Subtraction

Unit 3 · Fraction operation · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks:"New Thinking in Primary School Mathematics (Second Edition)" Volume 5A, Unit 3 + Modern Education 5A, Unit 10

Core Traps:□ T9 Fraction Operations - LCM Error Finding · The numerator forgets to expand simultaneously after the common division · The result is not reduced

SSPA Association: 🔴 High frequency The score test is required every year, accounting for about 15-20% of Paper 1

Prerequisite knowledge:P4 Addition and subtraction of fractions with the same denominator · LCM calculation · Equivalent fractions and extended fractions

Course goal:❶ Comparison of fractions with different denominators (common fraction → ratio) ❷ Addition of different denominators (three-step method for common denominators) ❸ Subtraction of different denominators ❹ Mixed application of addition and subtraction of fractions

Next class connection: Class 8 (In-depth practice of adding fractions with different denominators) - This class lays a solid foundation for comparison and basic addition and subtraction

Student Name:	Class:	Date:	Time Spent:

I.Warm-Up Questions (total 5 question · 5 minutes)

#	Question	Difficulty	Working Space (Show full working)
1	Calculate LCM(4, 6) = ?	Basic	
2	Calculate LCM(3, 5) = ?	Basic	
3	Compare $\frac{2}{3}$ and $\frac{3}{5}$ which one is bigger? Write down the judgment process.	Advanced	
4	Calculate $\frac{3}{7} + \frac{2}{7}$ = ? (Adding with the same denominator, P4 basics)	Basic	
5	Calculate $\frac{5}{9} - \frac{2}{9}$ = ? (Subtraction with the same denominator, P4 basic)	Basic	

II.Core Knowledge ＋ Worked Examples

Knowledge point 1: Comparison of fractions with different denominators - the ratio of the common denominator🔴 SSPA

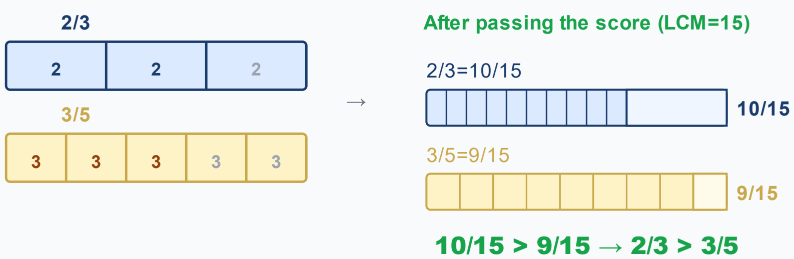
① Different denominators = different sizes of each part, the numerators cannot be directly compared

② **Three-step comparison method:**Find LCM → common denominator (expand the denominator to the same denominator) → compare the sizes of the molecules

③ After the common denominator, the fraction with a larger molecule is larger; the fraction with a smaller molecule is smaller

④ Comparison of three or more fractions: all common denominators to the same denominator (LCM of all denominators), and then arrange the molecules

⑤ **Trap T9:**LCM error finding → The basics of general division are wrong → The whole question is wrong!



Example 1

Comparing $\frac{3}{4}$ and $\frac{5}{6}$, which one is larger? Write down the general process.

Example 2 (Ordering three fractions)

Arrange $\frac{2}{3}$ 、 $\frac{3}{4}$ 、 $\frac{5}{6}$ from largest to smallest.

⚠ The most frequent error: compare the numerators directly without common denominators! Different denominators → different sizes of each portion → the numerators cannot be directly compared!

Knowledge Point 2: Addition of Fractions with Different Denominators - Three-Step Method for Common Fractions 🔴 SSPA Required Exam

① **Why can't you add it directly?**The denominator represents the serving size. Different denominators = different sizes of parts, you cannot add the numerator directly

② **Three steps:**Find LCM (least common multiple of the denominator) → Expand the fraction (the numerator is expanded simultaneously) → The denominator remains unchanged and the numerators are added

③ The answer must be**Reduction**(the simplest fraction); improper fraction must be**Convert mixed fractions**

④ **Trap T9:**When converting fractions, only change the denominator and forget to expand the numerator simultaneously!

①	②	③	④
Find LCM	**General/expanded**	**calculate**	**reduce**
Find the lowest common multiple using short division	The numerator and denominator change simultaneously to LCM	Adding numerators with the denominator unchanged	Simplest false fraction→mixed fraction

Example of trap detonation (the most important demonstration in this class)

calculate: $\frac{1}{2} + \frac{1}{3} = ?$

❌ Common mistakes (80% students)

$\frac{2}{5}$

Direct numerator plus numerator, denominator plus denominator

✅ Correct solution

$\frac{5}{6}$

LCM(2,3)=6 → $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$

Example 3

Calculation: $\frac{2}{3} + \frac{3}{4}$ = ? (Write down the four steps of ①LCM ②general split ③calculation ④reduction)

🧠 **Tip: "First understand the LCM, then multiply the numerator and keep the denominator unchanged. The answer must be simplified."**

⚠ The second most frequent error: only changing the denominator when dividing the whole number, forgetting to expand the numerator simultaneously. "The molecules follow and ride"!

⚠ The third most frequent error: the answer is not reduced. For example, 612 must be written as 12! One point will be deducted if the test is not simplified.

Knowledge point 3: Subtraction of fractions with different denominators - exactly the same steps as addition 🔴 SSPA required test

① Subtraction and addition use**exactly the same general division steps**- the only difference: the last step is "numerator subtraction" instead of "numerator addition"

② **Notes on subtraction:**Make sure that the numerator of the minuend ≧ the numerator of the minuend (after common division). If it is not enough → Borrowing is required (with fractions)

③ The answer also requires**Reduction**to the simplest fraction

Trap example (subtraction)

calculate: $\frac{5}{6} - \frac{1}{4} = ?$

❌ Common mistakes

$\frac{4}{2} = 2$

Direct numerator minus numerator, denominator minus denominator

✅ Correct solution

$\frac{7}{12}$

LCM(6,4)=12 → $\frac{10}{12} - \frac{3}{12} = \frac{7}{12}$

Example 4

calculate: $\frac{7}{8} - \frac{2}{3} = ?$

Knowledge point 4: Mixed addition and subtraction of fractions + application questions 🔴 SSPA must take the exam

① **Mixed addition and subtraction:**First divide all to the same denominator (LCM of all denominators), then calculate the numerator by +-, and finally reduce

② **Four steps to solve the application problem:**Circle the key words → Column formula (use + or − for judgment) → Calculate the common denominator → Write a complete answer

③ Keywords: "Total" "Total" = Addition; "remaining" and "difference" = subtraction; "more than..." → Find the number first and then add

Example 5

Calculation: $\frac{3}{4}+\frac{1}{6}-\frac{1}{3}=?$ (mixing of addition and subtraction of three fractions)

Example 6 (example of application questions)

A bottle of juice contains $\frac{3}{4}$ liters. Xiao Ming drank $\frac{1}{3}$ liters, and his mother poured in $\frac{1}{2}$ liters. How many liters of juice are there in the bottle now? (Answer with mixed fractions)

III. Lesson Layered Synchronization Practice

Basic layer (total 6 questions, everyone must do)

#	Question	Difficulty	Working Space
7	Comparing $\frac{1}{2}$ and $\frac{2}{5}$, which one is larger? (Write down the process of passing grades)	🌱	
8	calculate: $\frac{1}{3}+\frac{1}{6}=?$	🌱	
9	calculate: $\frac{1}{4}+\frac{1}{2}=?$	🌱	
10	calculate: $\frac{3}{5}-\frac{1}{10}=?$	🌱	
11	calculate: $\frac{2}{3}-\frac{1}{6}=?$	🌱	
12	Comparing $\frac{3}{8}$ and $\frac{5}{12}$, which one is larger? (LCM = 24)	🌿	

Advanced layer (total 5 questions, 🧗🚀 choose do)

#	Question	Difficulty	Working Space

13	calculate: $\frac{2}{5} + \frac{3}{10} = ?$	🌿	
14	calculate: $\frac{3}{4} - \frac{1}{6} = ?$	🌿	
15	calculate: $\frac{5}{6} - \frac{1}{3} = ?$	🌿	
16	Comparing $\frac{4}{7}$ and $\frac{5}{9}$, which one is larger? (Pass score required, LCM = 63)	🌿	
17	Calculation: $\frac{1}{2} + \frac{1}{3} - \frac{1}{6}$ = ? (mix of addition and subtraction)	🌿	

🌳 challenge layer (total 5 questions, 🚀 choose do, SSPAKiller Questions)

#	Question	Difficulty	Working Space
18	Sorting the three scores: Arrange $\frac{5}{6}$ 、 $\frac{3}{4}$ 、 $\frac{7}{12}$ from large to small.	🌳	
19	Calculation: $\frac{2}{3} + \frac{1}{4} + \frac{1}{6}$ = ? (Adding three fractions, LCM = 12)	🌳	
20	Calculation: $\frac{7}{8} - \frac{1}{6} - \frac{1}{4}$ = ? (Subtraction of three fractions)	🌳	
21	Xiao Ming originally had $\frac{3}{4}$ cakes, ate $\frac{1}{3}$ cakes, and bought $\frac{1}{2}$ cakes. How many cakes are there now?	🌳	
22	Among the three fractions |||SEP|||, what is the difference between the largest and the smallest? $\frac{2}{3}$ 、 $\frac{5}{12}$ 、 $\frac{1}{2}$, what is the difference between the maximum and minimum?	🌳	

IV.applicationquestionpractice (text questionspecial topic, SSPAMust-Test)

Application problem solving strategies 🔴 SSPA

① **Circle keywords:**"total" "total" → addition; "remaining" "difference" "less than" → subtraction

② **Column formula:**Use fraction symbols to write the complete formula

③ **Common fraction calculation:**Find LCM → Expand → Add and subtract molecules → Reduction

④ **Write an answer sentence:**"Answer:... (Unit)" - Step points will be deducted for not writing an answer sentence!

applicationquestion (total 6 questions, all must do, column → find a common denominator → calculate → answer sentence)

#	Question	Difficulty	Working Space
23	Xiao Mei drank $\frac{1}{4}$ liters of orange juice, and Xiao Ming drank $\frac{1}{2}$ liters. How many liters did they drink in total?	🌿	
24	A box of chocolate, my sister ate $\frac{2}{5}$ boxes, my brother ate $\frac{1}{3}$ boxes. How many boxes did the two of them eat in total?	🌿	
25	A rope is $\frac{5}{6}$ meters long, cut off $\frac{1}{4}$ meters. How many meters are left?	🌿	
26	The original water bottle contains $\frac{3}{4}$ liters, then pour $\frac{1}{3}$ liters into it. How many liters are there now? (Answer with mixed fractions)	🌳	
27	The cake shop sold $\frac{2}{5}$ cakes in the morning and $\frac{1}{3}$ cakes in the afternoon. How many were sold in total throughout the day?	🌿	
28	A bottle of juice contains $\frac{7}{8}$ liters. Xiao Ming drank $\frac{1}{4}$ liters and Xiaohua drank $\frac{1}{3}$ liters. How many liters are left? (subtract twice)	🌳	

V.⛰️ Ultimate challenge area (total 3 questions, 🚀 choose do, SSPA finale level)

#	Question	Difficulty	Working Space

⛰️ 1	Calculation: $\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{6}$ = ? (Adding four fractions with different denominators - find LCM(2,3,4,6))	⛰️	
⛰️ 2	Compare the equation A = $\frac{3}{4}+\frac{1}{6}$ with the equation B = |||SEP|||. Which result is greater? How much difference? $\frac{2}{3}+\frac{1}{4}$. Which result is greater? How much difference?	⛰️	
⛰️ 3	A rope is 2 meters long. $\frac{1}{3}$ was used on the first day, and the remaining $\frac{1}{2}$ was used on the second day. How many meters are left? (Hint: Remaining after the first day = $2-\frac{2}{3}=\frac{4}{3}$ meters; used the remaining on the second day $\frac{1}{2}=\frac{4}{3}\times\frac{1}{2}$)	⛰️	

VI.Class afterhomework

Basic must-do questions (total 5 questions, must write find a common denominator step)

#	Question	Difficulty	Working Space
H1	Comparing $\frac{2}{3}$ and $\frac{3}{4}$, which one is larger? Write down the general process.	🌱	
H2	calculate: $\frac{1}{2}+\frac{1}{6}=?$	🌱	
H3	calculate: $\frac{1}{3}+\frac{1}{9}=?$	🌱	
H4	calculate: $\frac{3}{4}-\frac{1}{2}=?$	🌱	
H5	calculate: $\frac{5}{6}-\frac{1}{3}=?$	🌱	

Advanced choose doquestion (total 3 questions, 🚀 choose do)

#	Question	Difficulty	Working Space
H6	calculate: $\frac{2}{3}+\frac{1}{4}=?$	🌿	
H7	One cake, Xiao Ming ate $\frac{1}{4}$ pieces, and Xiaohua ate $\frac{1}{3}$ pieces. How many cakes did the two of them eat together?	🌿	

H8	There are $\frac{5}{6}$ liters in a bucket of water, and $\frac{1}{4}$ liters are used. How many liters are left?	🌿	

VII. The Lessoncorecommon errorsummary

✅ **Self-examination in this hall (tick after completion)**

- ☐ I know the pitfalls of solving each knowledge point
- ☐ I can complete 🌱basic questions independently
- ☐ I can challenge 🌿advanced questions
- ☐ I remember the formula

🎯 **Review of Learning Objectives - After completing this lesson you should be able to:**

- ☐ Identify all trap types in our hall
- ☐ Solve 🌱basic questions independently (100% correct)
- ☐ Challenge 🌿Advanced questions (80%+ correct)
- ☐ Explain the lesson formula to classmates

✅ **本堂自我檢查（完成後打剔）**

- ☐ 我識得分辨每個知識點嘅陷阱
- ☐ 我能夠獨立完成🌱基礎題
- ☐ 我能夠挑戰🌿進階題
- ☐ 我記得住口訣

#	common error	Correct Approach
1	**Direct addition and subtraction with different denominators**：$\frac{1}{2}+\frac{1}{3}=\frac{2}{5}$	Different denominators → Be sure to connect the denominators first! Find LCM → Expand → Calculate again
2	**After splitting, the molecules forget to expand simultaneously.**	"Multiply the numerator" - multiply the denominator by the same number and multiply the numerator by the same number
3	**The answer is not reduced**：$\frac{6}{12}$ when the answer	The last step must be to check the common factors and convert them into the simplest fraction
4	**Improper fractions are not converted to mixed fractions**	Numerator > Denominator → Convert into mixed numbers (subject to the requirements of the division test)
5	**LCM error finding**(replace LCM with larger common multiple)	Short division confirmed to be the true "least" common multiple
6	**When comparing, only look at the numerator and not the denominator**	Different denominators → Different sizes of parts → Comparison must be made only after dividing the parts

7	**Missing sentences/units in application questions**	You must w rite "Answ er:..." + unit (liter, meter, unit), otherw ise step points w ill be deducted

🏠 家長角落 · Parent Corner

今日學咗咩？ 小朋友學咗**「Fractions with Different Denominators: Comparison, Addition and Subtraction」**。

最易錯嘅 3 個陷阱： 留意老師圈出嘅陷阱題目

你可以問小朋友：「你可唔可以解釋「Fractions with Different Denominators: Comparison, Addition and Subtraction」嘅最重要口訣俾我聽？」

溫馨提示：唔需要識教數學 — 只需要問小朋友「點解咁計？」同「有冇檢查陷阱？」就夠。

📝 **教師參考**：熱身題答案 → 1)___ 2)___ 3)___ 4)___ 5)___ | 課堂練習重點關注題號：🌳挑戰層 17-21